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Litterature:

H. H. Hamedani, *Probability and Statistics: The Science of Uncertainty*, 3rd ed., https://www.probabilitycourse.com

Course:

Sandsynlighedsteori og statistik (SWSTS-01)

Multiplication Principle

If all outcomes are equally likely, we compute the probability of an event A by:

$$P(A) = \frac{|A|}{|S|}$$

Counting Problem

Where:

- $|A|$: Number of favorable outcomes
- $|S|$: Total number of outcomes

Multiplication Principle

If a task consists of multiple stages:

- Stage 1 can be done in n_1 ways
- Stage 2 in n_2 ways
- ...
- Stage r in n_r ways

Each stage of a task is independent

Choices in each stage are independent

Then the total number of possible outcomes is:

$$n_1 \cdot n_2 \cdot \ldots \cdot n_r$$

Example: Configuring a Tablet

Choices:

- Screen size: Large or Small → 2 options
- Storage: 64GB, 128GB, 256GB → 3 options
- Color: Black or White → 2 options

Total combinations:

$$\mathbf{2 \cdot 3 \cdot 2 = 12 \quad \Rightarrow \quad P(\mathit{Large}, 128\ GB, \mathit{Black}) = \frac{1}{12}}$$

Multiplication Principle and probability

Example: Password and Hacker Guessing

Password Format:

- 2 lowercase letters (26 choices each)
- 1 uppercase letter (26 choices)
- 4 digits (10 choices each)

*e.g.: **ejT3018***

Number of possible passwords: $N = 26^3 \cdot 10^4$ by **multiplication principle**.

A hackerprogram generates 10^8 passwords guesses according to the above rule.

Scenario 1: Distinct Passwords (Duplicates not Allowed)

Assumption: All 10^8 guesses are different → outcomes are **disjoint** and **equally likely**.

$$P(one\ match) = \frac{number\ of\ guesses}{number\ of\ passwords} = \frac{10^8}{26^3 \cdot 10^4} \approx \mathbf{0.58}$$

Scenario 2: Independent passwords (Duplicates Allowed)

Assumption: Each of the 10^8 guesses is chosen **independently at random**, so **repeats are possible**.

$$P(at\ least\ one\ match) = 1 - \left(1 - \frac{1}{26^3 \cdot 10^4}\right)^{10^8} \approx \mathbf{0.43}$$

Multiplication Principle and probability

Example: Counting Subsets

Let A be a set with n elements, say:

$$A = \{a_1, a_2, \ldots, a_n\}$$

We want to determine **how many different subsets** can be formed from A, including the empty set and A itself.

For **each element** $a_i \in A$, we have exactly **2 options** when forming a subset:

- Include a_i
- Exclude a_i

These choices are made **independently** for each of the n elements. So, by the **multiplication principle**, the total number of distinct subsets is:

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n$$

Quiz

Let

$$A = \{a, b, c, d\}$$

A subset $S \subseteq A$ is chosen at random from the set of all subsets of A. What is the probability that

$$S = \{a, c\}?$$

A: 1/4

B: 1/8

C: 1/16

D: 2/16

Sampling

- **Sampling** means selecting elements from a set at random.
- Each element has an equal chance of being chosen.

With or Without Replacement

- **With replacement**:
  - After choosing an element, it is placed back into the set before the next draw.
  - This allows the same element to be chosen more than once.
  - Example:
    Set $A = \{a_1, a_2, a_3, a_4\}$
    Possible sample (3 draws with replacement): (a_3, a_1, a_3)
- **Without replacement**:
  - Once an element is chosen, it is removed from the set.
  - This means all chosen elements will be different.

Ordered or Unordered Sampling

- **Ordered sampling**:
  - The order of elements matters.
  - Example: $(a_1, a_2, a_3) \neq (a_2, a_3, a_1)$
- **Unordered sampling**:
  - The order of elements does not matter.
  - Example: $\{a_1, a_2, a_3\}$ is the same as $\{a_3, a_2, a_1\}$

Sampling and counting

Example: Ordered, With Replacement

How many way of picking 1 out 4 element from {A,B,C,D}

List:

(A), (B), (C), (D)

Total: 4 outcomes

How many way of picking 2 out 4 element from {A,B,C,D}

List:

(A, A), (A, B), (A, C), (A, D),

(B, A), (B, B), (B, C), (B, D),

(C, A), (C, B), (C, C), (C, D),

(D, A), (D, B), (D, C), (D, D)

Total: 16 outcomes

How many way of picking 3 out 4 element from {A,B,C,D}

List:

(A, A, A), (A, A, B), (A, A, C), (A, A, D),

(A, B, A), (A, B, B), (A, B, C), (A, B, D),

...

(D, D, C), (D, D, D)

Total: 64 outcomes

Quiz

How many way of picking 4 out 4 element from {A,B,C,D} in case of ordered, with replacement?

A: 128

B: 256

C: 512

D: 1024

Factorials

Definition:

The **factorial** of a positive integer n is written $n!$ and represents the product of all positive integers from 1 to n:

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1$$

Example:

Given a set $\{A, B, C\}$, how many possible orderings (permutations) are there?

Method 1:

List: (ABC), (ACB), (BAC), (BCA), (CAB), CBA) = **6 orderings**

Method 2:

3 options for the first position A or B or C.

2 options for the second position B or C (if A was picked before)

1 option for the third posistion C (if A and B was picked before)

Answer: $3! = 3 \cdot 2 \cdot 1 =$ **6 orderings**

Sampling and counting

Example: Ordered, Without Replacement

How many way of picking 1 out 4 element from {A,B,C,D}

Combination	Permutation 1
{A}	A
{B}	B
{C}	C
{D}	D

4 options for first position: So 4 in total

How many way of picking 2 out 4 element from {A,B,C,D}

Each has $2! = 2$ permutations.

Combination	Permutation 1	Permutation 2
{A, B}	(A, B)	(B, A)
{A, C}	(A, C)	(C, A)
{A, D}	(A, D)	(D, A)
{B, C}	(B, C)	(C, B)
{B, D}	(B, D)	(D, B)
{C, D}	(C, D)	(D, C)

Total ordered outcomes: $6 \times 2 = 12$.

Better method than counting directly: 4 options for first position, 3 options for second posistion so $4 \cdot (4 - 1) = 12$

Quiz

How many way of picking 3 out 4 element from {A,B,C,D} in case of ordered, without replacement?

A: 24

B: 48

C: 96

D: 192

Sampling and counting

Formula for Ordered, Without Replacement.

Pick k ordered elements out of n total elements:

- Stage 1 → n choices
- Stage 2 → $n-1$ choices
- Stage 3 → $n-2$ choices
- ...
- Stage k → $n-k+1$ choices

So by multiplication principle:

$$n(n-1)(n-2)\cdots(n-k+1).$$

Write:

$$n! = n(n-1)(n-2)\cdots 2\cdot 1.$$

To truncate after k factors, divide by the rest:

$$(n-k)! = (n-k)(n-k-1)\cdots 1.$$

So:

$$\boxed{P(n,k) = \frac{n!}{(n-k)!}}$$

Sampling and counting

Example: Unordered, Without Replacement

How many way of picking 1 out 4 element from {A,B,C,D}

Combination	Permutation 1
{A}	A
{B}	B
{C}	C
{D}	D

Total unordered outcomes: 4

How many way of picking 2 out 4 element from {A,B,C,D}

Each has $2! = 2$ permutations.

Combination	Permutation 1	Permutation 2
{A, B}	(A, B)	~~(B, A)~~
{A, C}	(A, C)	~~(C, A)~~
{A, D}	(A, D)	~~(D, A)~~
{B, C}	(B, C)	~~(C, B)~~
{B, D}	(B, D)	~~(D, B)~~
{C, D}	(C, D)	~~(D, C)~~

Total unordered outcomes: 6

So $\boldsymbol{Unordered\ without\ replacement = \frac{Ordered\ without\ replacement}{permutations} = \frac{12}{2!} = 6}$

Sampling and counting

Example: Unordered, Without Replacement

How many way of picking 3 out 4 element from {A,B,C,D}

Combination	Permutation 1	Permutation 2	Permutation 3	Permutation 4	Permutation 5	Permutation 6
{A, B, C}	(A, B, C)	(A, C, B)	(B, A, C)	(B, C, A)	(C, A, B)	(C, B, A)
{A, B, D}	(A, B, D)	(A, D, B)	(B, A, D)	(B, D, A)	(D, A, B)	(D, B, A)
{A, C, D}	(A, C, D)	(A, D, C)	(C, A, D)	(C, D, A)	(D, A, C)	(D, C, A)
{B, C, D}	(B, C, D)	(B, D, C)	(C, B, D)	(C, D, B)	(D, B, C)	(D, C, B)

Total unordered outcomes: 4

So $\boldsymbol{Unordered\ without\ replacement} = \frac{\boldsymbol{Ordered\ without\ replacement}}{\boldsymbol{permutations}} = \frac{\mathbf{24}}{\mathbf{3!}} = \mathbf{4}$

Quiz

How many way of picking 4 out 4 element from {A,B,C,D} in case of unordered, without replacement?

A: 0

B: 1

C: 2

D: 4

Sampling and counting

Formula for Unordered, Without Replacement.

Divide $P(n,k)$ with number of permutation

$$\frac{P(n,k)}{k!} = \frac{n!}{k!\,(n-k)!}$$

Several notations for this formula:

Combination notation:

$$C(n,k) = \frac{n!}{k!\,(n-k)!}$$

Binomial coefficient notation (most standard in mathematics):

$$\boxed{\binom{n}{k} = \frac{n!}{k!\,(n-k)!}}$$

Read as: "**n choose k**" or "the binomial coefficient."

Nb. Connection to The binomial theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Example: Unordered, With Replacement

How many way of picking 1 out 4 element from {A,B,C,D}

List:

(A), (B), (C), (D)

Total: 4 outcomes

How many way of picking 2 out 4 element from {A,B,C,D}

Multisets with repeats:

- $\{A, A\}$
- $\{B, B\}$
- $\{C, C\}$
- $\{D, D\}$

Multisets with two distinct items:

- $\{A, B\}$
- $\{A, C\}$
- $\{A, D\}$
- $\{B, C\}$
- $\{B, D\}$
- $\{C, D\}$

Total: 10 outcomes

Example: Unordered, With Replacement

How many way of picking 3 out 4 element from {A,B,C,D}

(A) All same

- $\{A, A, A\}$
- $\{B, B, B\}$
- $\{C, C, C\}$
- $\{D, D, D\}$

(B) Two same + one different

Two A's + one other

- $\{A, A, B\}$
- $\{A, A, C\}$
- $\{A, A, D\}$

Two B's + one other

- $\{A, B, B\}$
- $\{B, B, C\}$
- $\{B, B, D\}$

Two C's + one other

- $\{A, C, C\}$
- $\{B, C, C\}$
- $\{C, C, D\}$

Two D's + one other

- $\{A, D, D\}$
- $\{B, D, D\}$
- $\{C, D, D\}$

(C) All distinct

- $\{A, B, C\}$
- $\{A, B, D\}$
- $\{A, C, D\}$
- $\{B, C, D\}$

Total: 20 outcomes

Example: Unordered, With Replacement

Look for a numerical pattern

k	Count
1	4
2	10
3	20

We can find these numbers with binomial coefficients

$$\binom{4}{1} = 4, \quad \binom{5}{2} = 10, \quad \binom{6}{3} = 20.$$

Pattern suggests:

$$\binom{n+k-1}{k}.$$

Every **unordered sample with replacement** is determined by how many times each element appears, so counting samples is equivalent to counting **integer solutions** to

$$x_1 + \cdots + x_n = k.$$

Example:

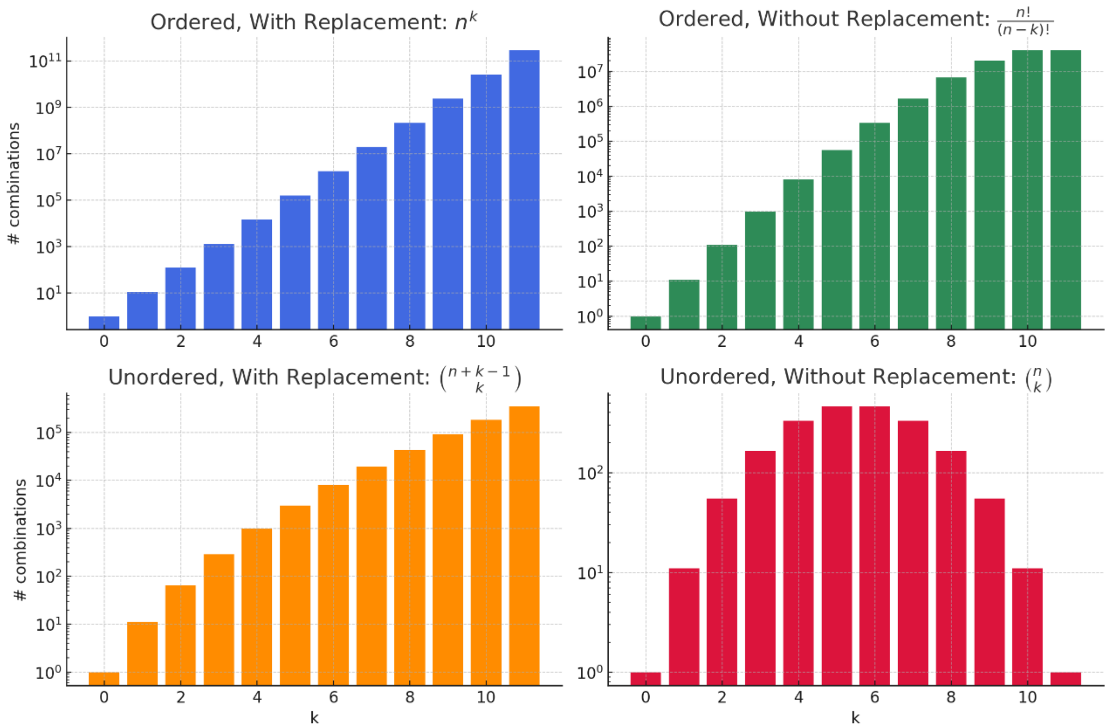
$$\{A, A, C\} \leftrightarrow (2, 0, 1, 0).$$

So this is a specific solution:

$$2 + 0 + 1 + 0 = 3$$

Combinatorics

Sampling Type	Formula	Example
Ordered, With Replacement	n^k = multiply n options k times	**PIN codes**: How many 3-digit codes using digits 0–9 (repeats allowed)? → $10^3 = 1000$
Ordered, Without Replacement	$\frac{n!}{(n-k)!}$ = arrange k out of n items without repeats $0 \le k \le n$	**Race medals**: 3 winners from 8 runners (gold, silver, bronze)? → $\frac{8!}{5!} = 336$
Unordered, Without Replacement	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ = number of k-item subsets from n items $0 \le k \le n$	**Committees**: Pick 3 students from 10 (order doesn't matter)? → $\binom{10}{3} = 120$
Unordered, With Replacement	$\binom{n+k-1}{k} = \frac{(n+k-1)!}{k!(n-1)!}$ = multiset combinations	**Pizza toppings**: Choose 3 toppings from 5 options (you can repeat a topping)? → $\binom{7}{3} = 35$



Example: Coin tosses

Consider tossing a coin 5 times.

What is the probability of getting 3 heads and 2 tails?

Method 1: count directly

There are $2^5 = 32$ possible configurations.

There are 10 possible configurations with 3 heads and 2 tails:

HHHTT, HHTHT, HHTTH, HTHHT, HTHTH, HTTHH, THHHT, THHTH, THTHH, TTHHH

So $P(3H, 2T) = \frac{10}{32} \approx 0.31$

Method 2: use combinatorics

A sequence with exactly 3 Heads is completely determined by which toss positions are Heads. Example: HTHHT ⟷ {1,3,4}.

If we choose these positions **in order**, we would get:

- First Head → 5 choices
- Second Head → 4 choices
- Third Head → 3 choices

$5 \cdot 4 \cdot 3 = 60$ permutations

But this **overcounts**, because the same set of positions can be chosen in different orders.

The 6 permutations (1,3,4), (1,4,3), (3,1,4), (3,4,1), (4,1,3), (4,3,1) all represent the same outcome.

So we divide to remove ordering: $\binom{5}{3} = \frac{5 \cdot 4 \cdot 3}{3!} = 10$

Quiz

You flip a fair coin 5 times.
What is the probability of getting **exactly 2 tails**?

A: 0.11

B: 0.31

C: 0.55

D: 0.67

Combinatorics and probability

Example: Cards

- From a standard 52-card deck, how many different 5-card poker hands can you draw?

  - You're choosing 5 cards
  - From 52 total
  - Without replacement
  - Order doesn't matter

$$\binom{52}{5} = \frac{52!}{5!(52-5)!} = 2{,}598{,}960$$

- In how many ways can you draw a **5-card poker hand** consisting of exactly **3 aces** and **2 tens**, from a standard 52-card deck?

Step 1: Choose 3 aces out of 4: $\binom{4}{3} = \frac{4!}{3!(4-3)!} = \frac{4!}{3!} = 4$

Step 2: Choose 2 tens out of 4: $\binom{4}{2} = \frac{4!}{2!(4-2)!} = \frac{4!}{2!\cdot 2!} = \frac{4\cdot 3}{2} = 6$

Step 3: Use the **Multiplication Principle:** $\binom{4}{3} \cdot \binom{4}{2} = 4 \cdot 6 = 24$

- **probability** of drawing exactly **3 aces and 2 tens** in a 5-card poker hand in a **single draw:**

$P(3A, 2\ tens) = \frac{24}{2{,}598{,}960} \approx 9.24 \cdot 10^{-6}$

Quiz

Suppose you draw 5 cards from a standard 52-card deck without replacement.
You want the probability of getting **exactly 3 aces and 2 tens**.

Can you compute this as:

$$P(3 \text{ aces and } 2 \text{ tens}) = P(3 \text{ aces}) \cdot P(2 \text{ tens})?$$

A. True — you can multiply because aces and tens are different card values.

B. False — because drawing cards without replacement introduces dependence between events.

C. False — because combinations cannot be multiplied.

D. False — because you need to use permutations, not probabilities.

Quiz

Suppose you draw 5 cards from a standard 52-card deck without replacement.
You want the probability of getting **exactly 3 aces and 2 tens**.

Can you compute this as:

$$P(3A, 2T) \;=\; P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1, A_2) \cdot P(T_4 \mid A_1, A_2, A_3) \cdot P(T_5 \mid A_1, A_2, A_3, T_4) \;?$$

A. False — because to handle "3 aces and 2 tens in any order," we must also account for the number of orderings, not just use the chain rule once.

B. False — because drawing cards without replacement makes events dependent, so multiplication is not valid.

C: True — because the chain rule works for dependent events as long as we condition properly.

D. True — because the chain rule applies only when the events are independent.

Quiz

You flip a fair coin **5 times**.

What is the probability of getting **more than 3 heads**?

A.

$$\frac{10}{32}$$

B.

$$\frac{5}{32}$$

C.

$$\frac{6}{32}$$

D.

$$\frac{16}{32}$$

Quiz

You flip a **biased coin** 6 times. The coin lands **heads with probability 0.6** and **tails with probability 0.4**.

Can you compute the probability of getting exactly 4 heads using:

$$P = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

A. False — because the total number of outcomes changes if the coin is unfair.

B. False — because 15 is no longer the number of favorable outcomes when the coin is biased.

C. False — because with a biased coin, outcomes are no longer equally likely, so classical probability doesn't apply.

D. True — you can still divide favorable outcomes by total outcomes.

Combinatorics and probability

Example: Biased Coin toss

You flip a **biased coin** 6 times.

The coin lands

- heads with probability 0.6,
- tails with probability 0.4.

What is the probability of getting **exactly 4 heads**?

Since all coin flips are **independent** from each other, we can write:

$$P(H \cap H \cap H \cap H \cap T \cap T) = P(H)^4 \cdot P(T)^2 = (0.6)^4 \cdot (0.4)^2$$

Each specific sequence with 4 H's and 2 T's (like **HHTTHH**) has:

$$P(\text{that sequence}) = (0.6)^4 \cdot (0.4)^2 = 0.1296 \cdot 0.16 = 0.020736$$

How many such sequences are there? Same as favorable outcomes from before:

$$\binom{6}{4} = \frac{6!}{4!\,(6-4)!} = \frac{6 \cdot 5}{2} = 15$$

Multiply:

$$P(exactly\ 4\ heads) = 15 \cdot 0.020736 \approx 0.31$$

Combinatorics and probability

Bernoulli Trial (Single Trial):

- A **Bernoulli trial** is an experiment with only **two outcomes**:
  - **1 = Success**, with probability p
  - **0 = Failure**, with probability $q = 1 - p$
- Common examples:
  - Coin toss: $\{\text{Head}, \text{Tail}\} = \{1, 0\}$
  - Digital signal: $\{1, 0\}$

Binomial Formula (Multiple Trials):

If you repeat a Bernoulli trial n times **independently**, the probability of getting exactly k successes is:

$$P(X = k) = \frac{n!}{k!(n-k)!} \cdot p^k (1-p)^{n-k}$$

- $\binom{n}{k}$: number of ways to place k successes in n trials
- p^k: probability of success repeated k times
- $(1-p)^{n-k}$: probability of failure repeated $n - k$ times

Combinatorics and probability

Example: Manufacturing Defects

A factory produces electronic components, and each unit has a **10% chance of being defective**.

You randomly sample **50 units**.

What is the probability that **exactly 15** of them are defective?

Given:

- $n = 50$ (number of trials/samples)
- $p = 0.1$ (probability of a defect = success)
- $k = 15$ (number of successes we want)

Use the **Binomial Formula**:

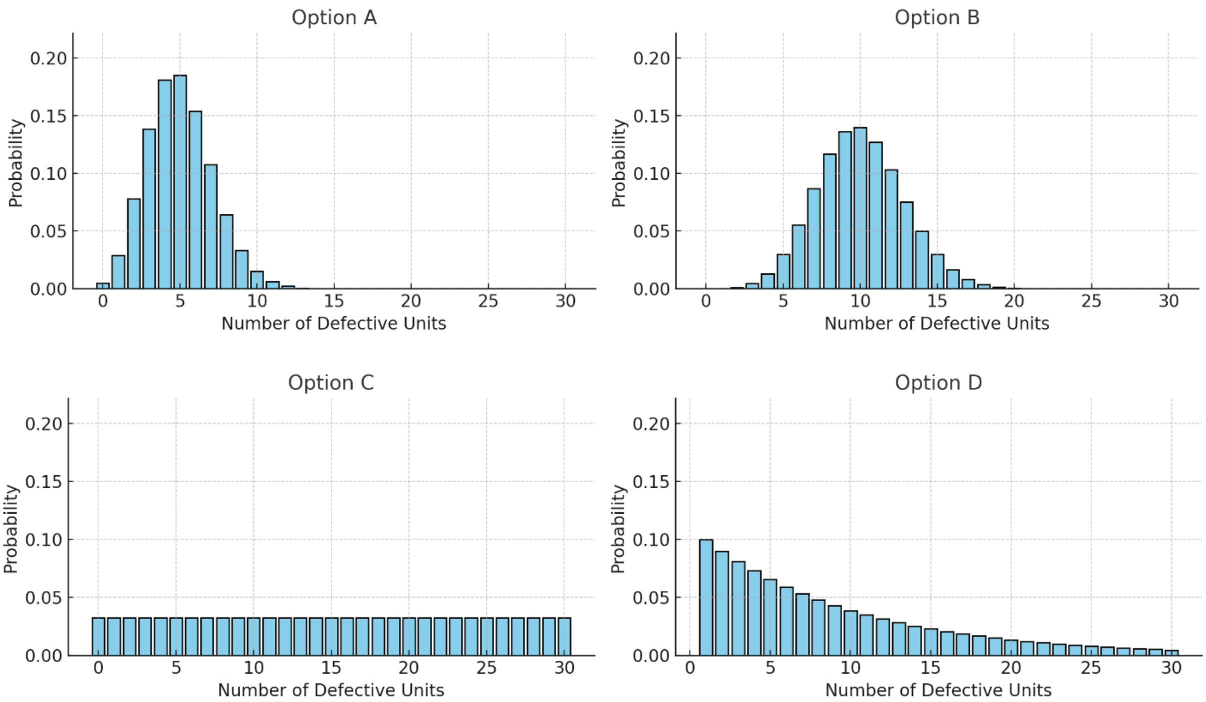
$$P(X = 15) = \frac{50!}{15!\,(50-15)!} \cdot (0.1)^{15} \cdot (0.9)^{35} \approx 5.63 \cdot 10^{-5}$$

Quiz

A factory produces electronic components, and each unit has a **10% chance of being defective**.

You randomly sample **50 units**.

Question: Which of the following distributions correctly represents the probability distribution for the number of defective units in your sample?



Stirling's Approximation (optional)

Sometimes we need approximations to handle big computations

Stirling's Approximation

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

Example: exact vs Stirling

$$Exact: 10! = 3.6288 \cdot 10^6$$

$$Stirling: \sqrt{2\pi \cdot 10} \cdot \left(\frac{10}{e}\right)^{10} = 3.5987 \cdot 10^6$$

$$Exact: 100! = 9.3326 \cdot 10^{157}$$

$$Stirling: \sqrt{2\pi \cdot 100} \cdot \left(\frac{100}{e}\right)^{100} = 9.3248 \cdot 10^{157}$$

$$Exact: 1000! = OverflowError$$

$$Stirling: \sqrt{2\pi \cdot 1000} \cdot \left(\frac{1000}{e}\right)^{1000} = 4.02354 \cdot 10^{2567}$$

References

Stirling's Approximation, Wikipedia:
https://en.wikipedia.org/wiki/Stirling%27s_approximation

Engelsk – dansk oversættelse

English	Dansk
Sampling	Udtagning
Ordered	Ordnet
Unordered	Uordnet
With replacement	Med tilbagelægning
Without replacement	Uden tilbagelægning
Multiplication principle	Multiplikationsprincippet
Factorial	Fakultet
Permutation	Permutation
Combination	Kombination
Binomial coefficient	Binomialkoefficient
Binomial formula	Binomialformel